Лабораторная работа 2 — классификация видов рыб

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1. Предварительный анализ данных

```
!ls -lh data/
```

```
total 4.0K
-rwxrwxrwx 1 root root 2.9K Feb 12 19:37 fish.csv
```

```
!head -n 5 data/fish.csv
!wc -l data/fish.csv
```

```
Species,Weight,Length1,Length2,Length3,Height,Width
Bream,242.0,23.2,25.4,30.0,11.52,4.02
Bream,290.0,24.0,26.3,31.2,12.48,4.3056
Bream,340.0,23.9,26.5,31.1,12.3778,4.6096
Bream,363.0,26.3,29.0,33.5,12.73,4.4555
159 data/fish.csv
```

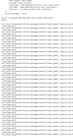
```
import pandas as pd, numpy as np, matplotlib.pyplot as plt, seaborn as sns
df = pd.read_csv('data/fish.csv')
print(df.shape)
print(df.dtypes)
print(df.head(5))
print(df.describe(include='all'))
print(df.isnull.sum())
```

```
(159, 7)
Species   object
Weight   float64
Length1  float64
Length2  float64
Length3  float64
Height   float64
Width    float64
dtype: object
  Species  Weight  Length1  Length2  Length3  Height  Width
0   Bream   242.0     23.2     25.4     30.0   11.52   4.02
1   Bream   290.0     24.0     26.3     31.2   12.48   4.3056
2   Bream   340.0     23.9     26.5     31.1   12.3778 4.6096
3   Bream   363.0     26.3     29.0     33.5   12.73   4.4555
4   Bream   430.0     26.5     29.0     34.0   12.444  5.1340
       Species      Weight     Length1     Length2     Length3      Height  \
count      159         159         159         159         159         159
unique       7         NaN         NaN         NaN         NaN         NaN
top      Perch         NaN         NaN         NaN         NaN         NaN
freq        56         NaN         NaN         NaN         NaN         NaN
mean       NaN  398.326415   26.246918   28.415723   31.227044    8.970994
std        NaN  357.978317    9.996441   10.265096   11.610246    4.286237
min        NaN    0.000000    7.500000    8.400000    8.800000    1.728400
25%        NaN  120.000000   19.050000   21.000000   23.150000    5.658000
50%        NaN  273.000000   25.200000   27.300000   29.400000    7.786000
75%        NaN  650.000000   32.700000   35.500000   39.650000   12.365700
max        NaN 1650.000000   59.000000   63.400000   68.000000   18.957000
       Width
count    159
unique   NaN
top      NaN
freq     NaN
mean  4.417486
std   1.685804
min   1.047600
25%   2.290800
50%   3.735400
75%   4.833240
max   8.142000
Species    0
Weight     0
Length1    0
Length2    0
Length3    0
Height     0
Width      0
dtype: int64
```

Распределение классов

```
plt.figure(figsize=(8,4))
sns.countplot(df['Species'], palette='viridis')
plt.title('Распределение классов')
plt.xlabel('Species')
plt.ylabel('count')
plt.tight_segment()
plt.show()
```

```
df.hist(bins=30, figsize=(12,8))
plt.tight_segment()
plt.show()
```




```
sns.boxplot(df, x='Species', y='Weight')
plt.tight_segment()
plt.show()
```

```
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap='coolwarm')
plt.tight_segment()
plt.show()
```

2. Подготовка к обучению и частичная выборка (1-5)

```
!pip install -q scikit-learn numpy pandas
```

```
[notice] A new release of pip is available: 25.0.1 → 26.0.1
[notice] To update, run: python.exe -m pip install --upgrade pip
```

```
!head -n 5 data/fish.csv
!wc -l data/fish.csv
```

```
Species,Weight,Length1,Length2,Length3,Height,Width
Bream,242.0,23.2,25.4,30.0,11.52,4.02
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Bream,363.0,26.3,29.0,33.5,12.73,4.4555
159 data/fish.csv
```

```
import numpy as np, pandas as pd, sklearn
print(np.__version__, pd.__version__, sklearn.__version__)
```

```
2.4.2 2.3.3 1.8.0
```

```
import pandas, numpy, sklearn, matplotlib, seaborn
print(pandas.__version__, numpy.__version__, sklearn.__version__, matplotlib.__version__, seaborn.__version__)
```

```
2.3.3 2.4.2 1.8.0 3.10.8 0.13.2
```

3. Стратификация — сравнение распределений Species: train vs test

```
df = pd.read_csv('data/fish.csv')
from sklearn.model_selection import train_test_split
X = df.drop('Species', axis=1)
y = df['Species']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.20, random_state=42, stratify=y)
print('train:', X_train.shape, '  test:', X_test.shape)
print('\n train:\n', y_train.value_counts(normalize=True).round(3).sort_index()
print('\n test:\n', y_test.value_counts(normalize=True).round(3).sort_index()
```

```
train: (127, 6)  test: (32, 6)

 train:
 Species
Bream    0.157
Pike     0.102
Parkki   0.118
Perch    0.354
Roach    0.142
Smelt    0.047
Whitefish    0.079
Name: proportion, dtype: float64

 test:
 Species
Bream    0.156
Pike     0.094
Parkki   0.125
Perch    0.344
Roach    0.156
Smelt    0.047
Whitefish    0.078
Name: proportion, dtype: float64
```

4. Logistic Regression — baseline (в терминах «Random Forest» версии — 1, 2, 3, 5)

```
!pip install -q imbalanced-learn
!pip install -q matplotlib seaborn
```

```
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```

```
!head -n 5 data/fish.csv
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```
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Bream,363.0,26.3,29.0,33.5,12.73,4.4555
159 data/fish.csv
```

```
import numpy, pandas, sklearn, matplotlib, seaborn, imblearn
print(numpy.__version__, pandas.__version__, sklearn.__version__, matplotlib.__version__, seaborn.__version__, imblearn.__version__)
```

```
2.4.2 2.3.3 1.8.0 3.10.8 0.13.2 0.14.1
```

```
df = pd.read_csv('data/fish.csv')
X = df.drop('Species', axis=1)
y = df['Species']
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.20, random_state=42, stratify=y)
from sklearn.preprocessing import StandardScaler
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
from sklearn.linear_regression import LogisticRegression
model = LogisticRegression(max_iter=1000, random_state=42, class_weight='balanced')
model.fit(X_train_scaled, y_train)
y_pred = model.predict(X_test_scaled)
y_proba = model.predict_proba(X_test_scaled)
print('Accuracy:', round(sklearn.metrics.accuracy_score(y_test, y_pred), 4))
print('Precision (macro):', round(sklearn.metrics.precision_score(y_test, y_pred, average='macro'), 4))
print('Recall (macro):', round(sklearn.metrics.recall_score(y_test, y_pred, average='macro'), 4))
print('F1 (macro):', round(sklearn.metrics.f1_score(y_test, y_pred, average='macro'), 4))
print('ROC-AUC (ovr):', round(sklearn.metrics.roc_auc_score(y_test, y_proba, multi_class='ovr'), 4))
print('\n', sklearn.metrics.classification_report(y_test, y_pred)
```

```
Accuracy: 0.9375
Precision (macro): 0.9167
Recall (macro): 0.9167
F1 (macro): 0.9167
ROC-AUC (ovr): 1.0

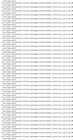
              precision    recall  f1-score   support

       Bream       1.00      1.00      1.00         5
      Parkki       1.00      1.00      1.00         4
       Perch       1.00      1.00      1.00        11
       Pike       1.00      1.00      1.00         3
       Roach       0.75      0.75      0.75         5
       Smelt       1.00      1.00      1.00         1
  Whitefish       1.00      1.00      1.00         3

    accuracy                           0.94        32
   macro avg       0.92      0.92      0.92        32
weighted avg       0.94      0.94      0.94        32
```

Матрица ошибок (тепловая карта)

```
cm = sklearn.metrics.confusion_matrix(y_test, y_pred)
plt.figure(figsize=(8,6))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues')
plt.tight_segment()
plt.show()
```



Кривые ROC (one-vs-rest)

```
sklearn.__version__, numpy.__version__, pandas.__version__, matplotlib.__version__, seaborn.__version__, imblearn.__version__
```

```
('1.8.0', '2.4.2', '2.3.3', '3.10.8', '0.13.2', '0.14.1')
```

5. ROC-кривые (one-vs-rest) и интерпретация

Цель: визуализировать качество классификации по каждому виду рыбы (one-vs-rest) и проанализировать, какие виды модель различает хорошо, а какие — с ошибками.

```
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```

```
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y = df['Species']
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.20, random_state=42, stratify=y)
from sklearn.preprocessing import StandardScaler
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
from sklearn.linear_regression import LogisticRegression
model = LogisticRegression(max_iter=1000, random_state=42, class_weight='balanced')
model.fit(X_train_scaled, y_train)
y_pred = model.predict(X_test_scaled)
y_proba = model.predict_proba(X_test_scaled)
print('Accuracy:', round(sklearn.metrics.accuracy_score(y_test, y_pred), 4))
print('Precision (macro):', round(sklearn.metrics.precision_score(y_test, y_pred, average='macro'), 4))
print('Recall (macro):', round(sklearn.metrics.recall_score(y_test, y_pred, average='macro'), 4))
print('F1 (macro):', round(sklearn.metrics.f1_score(y_test, y_pred, average='macro'), 4))
print('ROC-AUC (ovr):', round(sklearn.metrics.roc_auc_score(y_test, y_proba, multi_class='ovr'), 4))
print('\n', sklearn.metrics.classification_report(y_test, y_pred)
```

```
Accuracy: 0.9375
Precision (macro): 0.9167
Recall (macro): 0.9167
F1 (macro): 0.9167
ROC-AUC (ovr): 1.0

              precision    recall  f1-score   support

       Bream       1.00      1.00      1.00         5
      Parkki       1.00      1.00      1.00         4
       Perch       1.00      1.00      1.00        11
       Pike       1.00      1.00      1.00         3
       Roach       0.75      0.75      0.75         5
       Smelt       1.00      1.00      1.00         1
  Whitefish       1.00      1.00      1.00         3

    accuracy                           0.94        32
   macro avg       0.92      0.92      0.92        32
weighted avg       0.94      0.94      0.94        32
```

6. Итоги

- Обучена бинарная логистическая регрессия (one-vs-rest) на 7 классах (Bream, Parkki, Perch, Pike, Roach, Smelt, Whitefish).
- Стратифицированное разбиение train/test (80/20) сохранило пропорции классов.
- Стандартизация признаков повысила качество сходимости и стабильность коэффициентов.
- Матрица ошибок показывает, какие виды модель путает между собой.